1. Cofactor-projected affine base on an ordinary binary curve

2026-10-17 · Input construction and independent Sage replay; no index-calculus speed claim.

The exact fixture is $y^2 + xy = x^3 + 31$ over $\mathrm{GF}(2^{61})$ with field polynomial $z^{61} + z^5 + z^2 + z + 1$ and group order $4r$, where $r = 576460752557211763$ is prime. The target subgroup has 60-bit order. The fixture certificate, producer, input record, and Sage replay are linked in the companion Markdown report.

1.1. Sparse raw base

Let $V = \mathrm{span}_{F_2}\{1, z, ..., z^8\}$. The linearized polynomial $L_{V(X)} = \prod_{v \in V}(X + v)$ has exact kernel V. For each predetermined offset c in $\{512, 1024, 1536, 2048\}$, the raw decomposition points have liftable x coordinates satisfying $L_{V(x)} = L_{V(c)}$. The polynomial has 10 nonzero Frobenius terms, plus the affine constant. Each placement scans 512 x values. The selected offset minimizes the gap to the default $m = 8$ base size $B = 287$ before target selection.

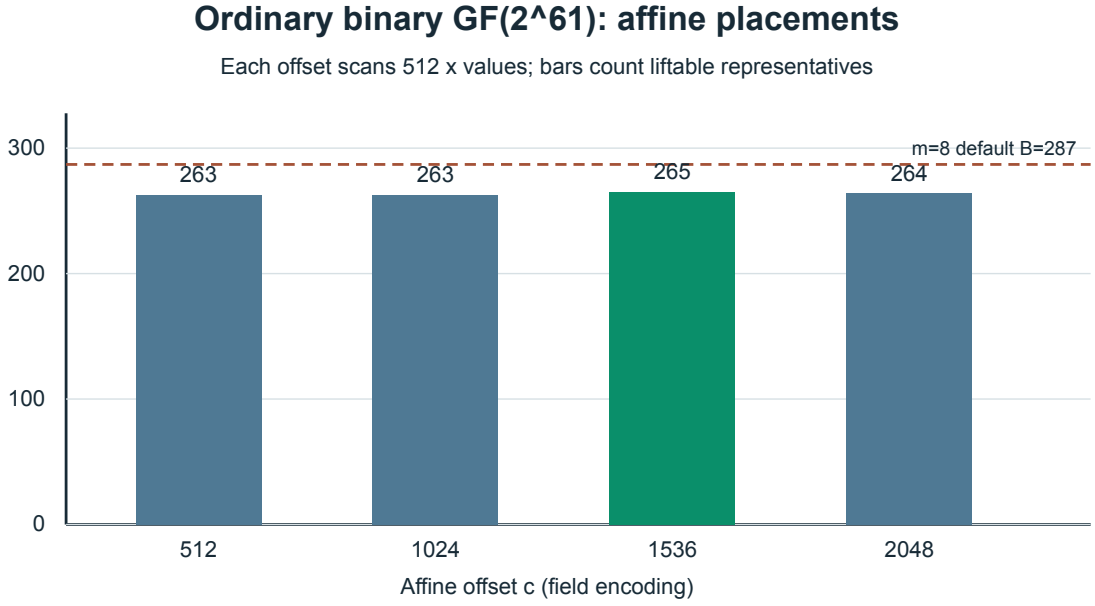


Figure 1: Measured realized base representatives at four predetermined offsets; all counts were independently replayed in Sage. These are input counts, not solver outcomes.

The counts are 263, 263, 265, and 264; $c = 1536$ is selected. An equal-size seeded random control needed 517 draws to retain 265 liftable x values. Preparation recorded 9.642826 s wall, 5.599027 s CPU, and 69,500,928 bytes peak RSS on a shared host. The timings describe this one preparation process only.

1.2. Exact projection and target channels

For a raw point P_i, the log column is $F_i = [4]P_i$ in the order-r subgroup. For target R, let $R_0 = [4^{-1} \bmod r]R$ and choose order-four $T = [r]P_{\mathrm{source}}$. Its four raw preimages are $R_0 + jT$, $j \in \{0, 1, 2, 3\}$. Therefore a raw equality $\sum_i P_i = R_0 + jT$ implies $\sum_i F_i = R$. The independent replay checked all four channels, one public scalar-17 target, and a planted three-point sum. The four-channel work remains unpriced.

The sparse predicate applies to raw x coordinates; subgroup columns are projected points. A future solver formulation must connect these two representations and include curve lifts, signs, torsion channels, relation yield, useful rank, linear algebra, and target work. No degree-of-regularity or complete ECDLP measurement exists for this construction. Direct S_8 has seven summands plus target; eight summands require S_9.

1.3. Evidence boundary

Producer: `tools/icperf_binary_projected_base.py`; input: `research/icperf_binary_projected_base_20261017/base.json`; independent replay: `research/icperf_binary_projected_base_20261017/sage_replay.json`. The full-width arity-grid problem remains open. Canonical frontier graphs were checked and left unchanged because this is an input representation result.